

Lecture 19

30th Oct

Algorithm BFS (G, v)

Input : $G = (V, E)$ and $v \in V$

Output :

begin

mark v ; put v in a queue;

while the queue is not empty do

remove the first vertex w from queue;

perform pre work;

for all edges (w, x) s.t. x is unmarked do

mark x ; add (w, x) to T ,

put x in queue;

end

Algo to find GCD of two numbers

$$O(\log(\max(a, b)))$$

Polynomial ,

Homer's Rule

begin

$p := a_n ;$

for $i = 1$ to n do

$p := x + p + a_{n-i}$

end

begin

$p \leftarrow a_n$

for $i = 1$ to $n-1$ do

$p \leftarrow a_i + n_0 + p$

end for

return p

end

$$A_{m \times n} \times B_{n \times n}$$

$$A_i = \sum_{k=1}^{n/2} a_{i, 2k-1} \cdot a_{i, 2k}$$

$$B_j = \sum_{k=1}^{n/2} b_{2k-1, j} \cdot b_{2k, j}$$

$$C_{ij} = \sum_{k=1}^{n/2} (a_{i, 2k-1} \cdot b_{2k, j} + a_{i, 2k} \cdot b_{2k-1, j})$$

$$- A_i - B_j$$

Winograd's Algorithm

Strassen's Algorithm